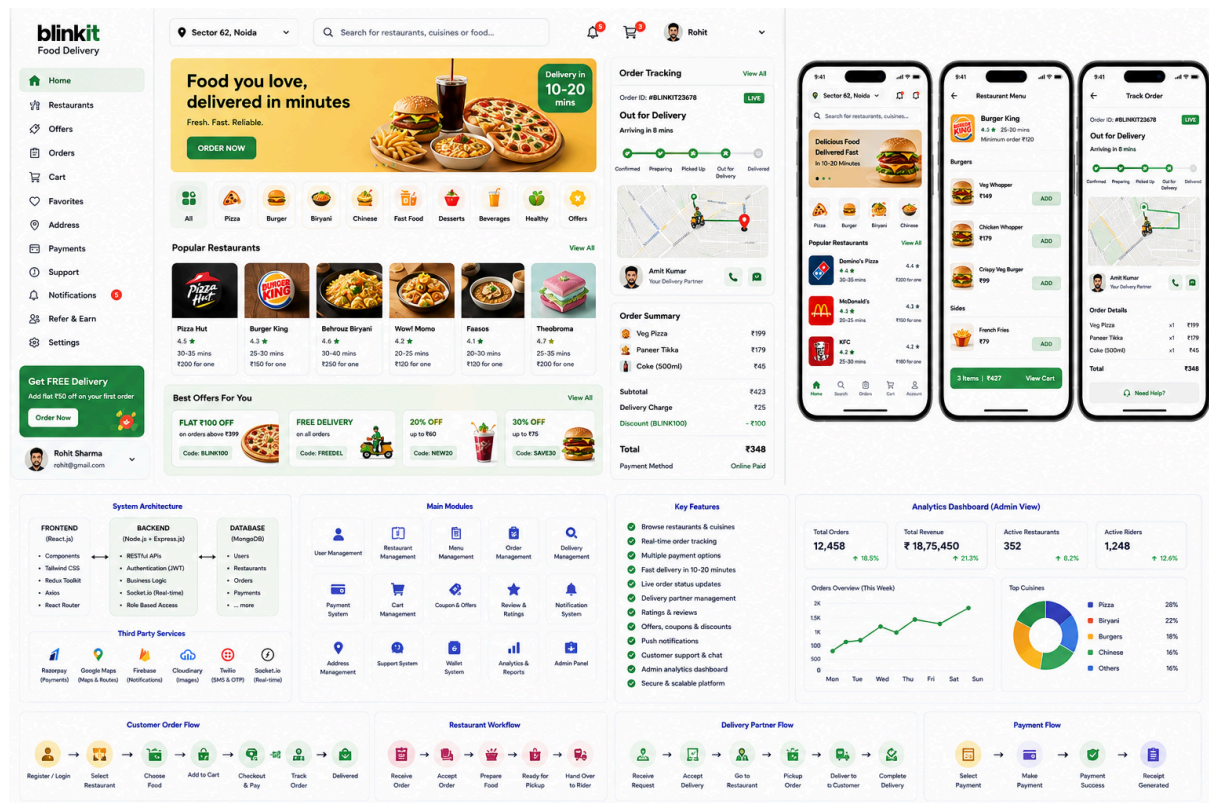


Blinkit Food Delivery Platform



1. Project Introduction

What is Blinkit Food Delivery?

Blinkit Food Delivery is a hyperlocal food delivery platform designed to deliver food, snacks, beverages, bakery products, ready-to-eat meals, and daily essentials from nearby restaurants, cloud kitchens, cafes, and food outlets within minutes.

Unlike traditional food delivery platforms where delivery can take 30–60 minutes, Blinkit Food Delivery focuses on:

- Ultra-fast delivery
- Nearby food partners
- Real-time inventory
- Smart order routing
- Instant customer experience

The platform connects:

- Customers
- Restaurants
- Cloud Kitchens
- Delivery Partners

- Admins

through a centralized ecosystem that manages food ordering, payments, delivery logistics, and customer engagement.

2. Business Problem

Modern customers expect food delivery to be:

- Fast
- Convenient
- Contactless
- Trackable

However, traditional systems face several issues.

Customer Problems

- Long waiting times
 - Order tracking issues
 - Payment failures
 - Poor delivery visibility
-

Restaurant Problems

- Order management complexity
 - Peak-hour overload
 - Inventory visibility challenges
-

Delivery Partner Problems

- Poor route optimization
 - Multiple delivery coordination issues
-

Business Problems

- Low customer retention
- Inefficient dispatch systems

- Operational inefficiencies
-

The platform solves these challenges through technology-driven food delivery management.

3. Project Objective

The platform should allow:

Customers

- Browse restaurants
 - Order food
 - Make payments
 - Track deliveries
-

Restaurants

- Manage menus
 - Process orders
 - Track revenue
-

Delivery Partners

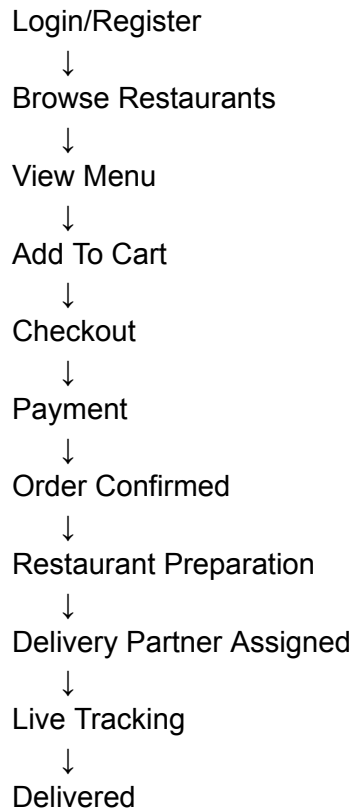
- Accept deliveries
 - Navigate routes
 - Manage earnings
-

Admin

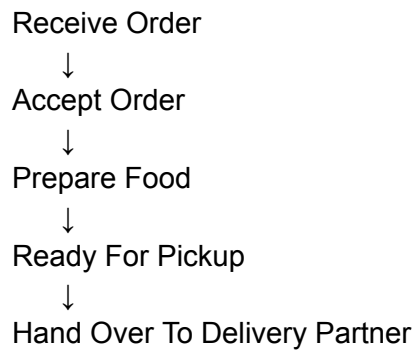
- Monitor platform
 - Manage restaurants
 - Manage deliveries
 - Generate reports
-

4. Real World Business Workflow

Customer Journey



Restaurant Journey



Delivery Partner Journey



↓
Complete Delivery

5. User Roles

Customer

Primary food ordering user.

Responsibilities

- Search food
 - Place orders
 - Manage addresses
 - Track deliveries
 - Submit reviews
-

Restaurant Owner

Responsible for food preparation.

Responsibilities

- Manage menu
 - Manage orders
 - Track revenue
 - Update food availability
-

Delivery Partner

Responsible for deliveries.

Responsibilities

- Accept orders
 - Pickup food
 - Deliver food
 - Manage earnings
-

Admin

Responsible for platform management.

Responsibilities

- Manage users
 - Manage restaurants
 - Monitor deliveries
 - View analytics
-

6. Core Functional Modules

Module 1: Authentication & User Management

Purpose:

Provide secure access.

Features

- Registration
 - Login
 - OTP Verification
 - Forgot Password
 - JWT Authentication
 - Role-Based Access Control
-

Module 2: Restaurant Management

Purpose:

Manage restaurant onboarding.

Restaurant Information

Restaurant Name

Owner Name

Address

Cuisine Type

Opening Hours

GST Number

Contact Details

Restaurant Categories

Cafe

Pizza

Bakery

Fast Food

South Indian

North Indian

Chinese

Desserts

Module 3: Menu Management

Purpose:

Manage food products.

Food Item Information

Food Name

Description

Price

Category

Food Type

Image

Availability

Example

Veg Burger

₹149

Fast Food

Available

Food Categories

Pizza

Burger

Momos

Biryani

Coffee

Tea

Juices

Desserts

Module 4: Food Discovery System

Purpose:

Help customers find food quickly.

Search Features

Restaurant Search

Food Search

Cuisine Search

Filters

Veg

Non-Veg

Price

Rating

Delivery Time

Offers

Sorting

Popularity

Fastest Delivery

Highest Rated

Lowest Cost

Module 5: Cart Management

Purpose:

Manage food selections.

Features

- Add To Cart

- Remove Item
- Quantity Management
- Apply Coupon

Cart Summary

Subtotal

GST

Discount

Delivery Charges

Final Amount

Module 6: Order Management

The most important module.

Purpose:

Manage complete food ordering lifecycle.

Order Workflow

Placed



Accepted



Preparing



Ready For Pickup



Picked Up



Delivered

Order Information

Order Number

Customer

Restaurant

Items

Amount

Status

Module 7: Delivery Management

Purpose:

Manage food deliveries.

Features

- Delivery Assignment
 - Route Planning
 - Delivery Tracking
 - Delivery History
-

Delivery Status

Assigned

Accepted

Picked Up

On The Way

Delivered

Module 8: Real-Time Order Tracking

Purpose:

Provide live delivery visibility.

Tracking Information

Restaurant Accepted

Preparing Food

Delivery Assigned

Picked Up

Near You

Delivered

Advanced Features

Live Map

ETA

Driver Information

Module 9: Address Management

Purpose:

Manage delivery addresses.

Address Types

Home

Hostel

Office

PG

Address Information

Name

Phone

Address

City

Pincode

Module 10: Payment Management

Purpose:

Handle financial transactions.

Payment Methods

UPI

Credit Card

Debit Card

Net Banking

Wallet

Cash On Delivery

Features

- Payment Verification
 - Refund Processing
 - Invoice Generation
 - Payment History
-

Module 11: Coupon & Offers Management

Purpose:

Increase sales.

Examples

WELCOME100

FREEDELIVERY

SAVE50

BLINKIT20

Features

- Coupon Validation
 - Restaurant Offers
 - Cashback Programs
-

Module 12: Ratings & Reviews

Purpose:

Improve quality and trust.

Customers Can Review

Food Quality

Taste

Packaging

Delivery Experience

Example

★★★★★

Fresh Food

Fast Delivery

Module 13: Notification System

Purpose:

Keep users updated.

Notifications

Order Confirmed

Preparing Food

Out For Delivery

Delivered

Offer Alerts

Module 14: Customer Support System

Purpose:

Handle complaints and refunds.

Features

Raise Ticket

Refund Request

Late Delivery Complaint

Wrong Item Complaint

Live Chat Support

Module 15: Restaurant Analytics

Purpose:

Help restaurants grow.

Analytics

Daily Orders

Revenue

Top Selling Items

Peak Hours

Customer Ratings

Module 16: Delivery Partner Analytics

Purpose:

Track performance.

Analytics

Completed Deliveries

Average Delivery Time

Earnings

Acceptance Rate

Module 17: Admin Analytics Dashboard

Purpose:

Provide business insights.

Platform Analytics

Total Customers

Total Restaurants

Active Delivery Partners

Orders Today

Revenue

Top Categories

7. Frontend Screen Planning

Customer Screens

Home

Restaurants

Restaurant Details

Food Details

Cart

Checkout

Orders

Live Tracking

Addresses

Profile

Support

Restaurant Screens

Dashboard

Orders

Menu

Inventory

Analytics

Offers

Delivery Partner Screens

Dashboard

Assigned Orders

Maps

Earnings

History

Admin Screens

Dashboard

Users

Restaurants

Orders

Payments

Reports

Settings

8. Database Design

Core Collections

users

restaurants

menuitems

categories

carts

orders

deliveries

payments

addresses

coupons

reviews

notifications

tickets

9. Business Relationships

Customer

|
|---- Orders
|
|---- Addresses

Restaurant

|
|---- Menu Items
|
|---- Orders

Order

|
|---- Payment
|
|---- Delivery

Delivery Partner

|
|---- Deliveries

10. Third-Party Integrations

Google Maps API

Purpose:

- Live Tracking
 - Route Optimization
 - ETA Calculation
-

Razorpay

Purpose:

- Online Payments
-

Firebase

Purpose:

- Push Notifications
-

Socket.IO

Purpose:

- Real-Time Tracking
 - Real-Time Order Updates
-

Cloudinary

Purpose:

- Food Images
 - Restaurant Images
-

Twilio

Purpose:

- OTP Verification
- SMS Alerts

11. Future Enhancements

AI Food Recommendation

Recommend food based on ordering history.

AI Delivery Routing

Optimize delivery assignments.

Subscription Membership

Premium delivery membership.

Voice Ordering

Order food using voice commands.

Group Ordering

Multiple students or employees can place a shared order.

Smart Kitchen Forecasting

Predict demand based on previous orders.

Drone Delivery Support

Future logistics enhancement.

Expected Outcome

The completed Blinkit Food Delivery Platform should function as a full-scale hyperlocal food delivery ecosystem where:

- Customers receive food quickly from nearby restaurants.
- Restaurants efficiently manage menus and orders.
- Delivery partners perform optimized deliveries.
- Admins monitor the entire business operation.

This project provides students with experience in food commerce, order lifecycle management, payment gateways, geolocation services, real-time tracking, coupon engines, notifications, analytics dashboards, customer support workflows, and enterprise-level MERN architecture, making it one of the strongest food-tech projects for portfolio development.